

Conditional half-exponent reduction for the two-point metric constant

Run fa_banach_001, lane 0

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Status. This is a conditional packet. It isolates one explicit scalar inequality whose proof would give a negative answer to Question 3 of Cuth–Raunig [1]. The packet does not claim that the scalar inequality has been proved.

1 Source question

The source paper is Marek Cuth and Tomas Raunig, *Canonical embedding of Lipschitz-free p -spaces*, arXiv:2311.02920 [1]. In Section 4, after Proposition 3.24, the authors ask:

Is it true that $\mathbf{aa}_p^1(2) > 1$ for every $p \in (0, 1)$?

The crop below shows the source context.

4. OPEN QUESTIONS

The most important problem remains [2] Question 6.2], that is, whether given p -metric spaces $\mathcal{N} \subset \mathcal{M}$, $\mathcal{N}$ is p -amenable in $\mathcal{M}$ (Theorem [3.21] gives positive answer for metric spaces). By [1] Lemma 2.5], it is equivalent to ask whether we have $\sup_{n \in \mathbb{N}} \mathbf{aa}_p(n) < \infty$. Even after a considerable effort (including numerical simulation, where we considered several millions of randomly generated p -metric spaces) we were not able to find a single example of p -metric spaces $\mathcal{N} \subset \mathcal{M}$ for which we would have $\mathbf{a}_p(\mathcal{N}, \mathcal{M}) > 2^{1/p}$. This leads us to ask the following.

Question 1. Given $0 < p \leq q \leq 1$, is it true that $\sup_{n \in \mathbb{N}} \mathbf{aa}_p^q(n) \leq 2^{1/q}$? Is it true at least for $q = p$ or for $q = 1$?

Concerning the value of $\mathbf{aa}_p(2)$ the best estimate we have is contained in Proposition [3.24]. This leads us to ask e.g. the following two related problems, which seem to be of a very technical nature. We suspect the answer to the first one is positive, and the answer to the second one is negative (which would be interesting as it would imply that the problem of whether $\mathcal{N}$ is isometrically p -amenable in $\mathcal{M}$ for metric spaces $\mathcal{N} \subset \mathcal{M}$ with $|\mathcal{N}| = 3$, depends on the value of $p \in (0, 1)$).

Question 2. Is it true that $\mathbf{aa}_p(2) = \frac{4}{4+2^p(2^p-2)}$ for every $p \in (0, 1)$?

Question 3. Is it true that $\mathbf{aa}_p^1(2) > 1$ for every $p \in (0, 1)$?

REFERENCES

2 Definitions and reduction used

For $0 < p \leq q \leq 1$, $\text{aa}_p^q(2)$ denotes the supremal p -amenability constant over two non-base-point q -metric spaces $\mathcal{N} \subset \mathcal{M}$. The source paper's Theorem 3.17 reduces $\text{aa}_p^q(2)$ to weighted rooted trees with two leaves. For $q = 1$, the only nontrivial tree has vertices $0, z, x, y$, edges $0z, zx, zy$, and positive edge lengths u, v, w . Thus

$$d(0, x) = u + v, \quad d(0, y) = u + w, \quad d(x, y) = v + w.$$

For coefficients $a, b \in \mathbb{R}$, put

$$X = |a|^p, \quad Y = |b|^p, \quad Z = |a + b|^p.$$

Corollary 2.4 of [1] gives the p -th power of the norm in $\mathcal{F}_p(\{0, x, y\})$ as the minimum of

$$\begin{aligned} N_1 &= X(u + v)^p + Y(u + w)^p, \\ N_2 &= Z(u + v)^p + Y(v + w)^p, \\ N_3 &= Z(u + w)^p + X(v + w)^p. \end{aligned}$$

The corresponding tree expression in $\mathcal{F}_p(\{0, z, x, y\})$ is

$$D = Zu^p + Xv^p + Yw^p.$$

Therefore $\text{amen}_p(\{0, x, y\}, \{0, z, x, y\}) \leq 1$ follows from

$$\min\{N_1, N_2, N_3\} \leq D$$

for all a, b, u, v, w .

3 Proof intuition

At $p = 1/2$, the tree distances raised to the p -power are Euclidean distances in the Hilbert embedding of a weighted tree: each tree edge becomes an orthogonal vector of length equal to the square root of the edge length. This makes the two-leaf problem a four-point Euclidean tripod inequality.

The certificate below averages the three possible three-point tree formulas N_1, N_2, N_3 . The weights are chosen so that the X - and Y -coefficients match the fork expression D exactly. The only remaining issue is the Z -coefficient, and that leftover condition is precisely Scalar Lemma E.

4 Conditional dependency

Conditional Lemma 1 (Scalar Lemma E). *For all $u, v, w > 0$, set*

$$\begin{aligned} A &= \sqrt{u + v}, & B &= \sqrt{u + w}, & C &= \sqrt{v + w}, \\ U &= \sqrt{u}, & V &= \sqrt{v}, & W &= \sqrt{w}. \end{aligned}$$

Then

$$2AB(V + W - C) \geq (A + B - C)(AW + BV - UC). \tag{E}$$

Numerical evidence for Lemma 1 is included in `code/check_half_tripod_certificate.py`. A useful equivalent Euclidean form is obtained by writing $U = r$, $V = x$, $W = y$. Then

$$A = \sqrt{r^2 + x^2}, \quad B = \sqrt{r^2 + y^2}, \quad C = \sqrt{x^2 + y^2},$$

and (E) becomes

$$2AB(x + y - C) \geq (A + B - C)(Ay + Bx - rC). \quad (\text{E}')$$

After normalizing $x^2 + y^2 = 1$ and putting $m = xy$, this is the two-variable inequality

$$\begin{aligned} 2P(\sqrt{1 + 2m} - 1) \\ \geq (\sqrt{2r^2 + 1 + 2P} - 1)(\sqrt{r^2 + 2m^2 + 2mP} - r), \end{aligned}$$

where $P = \sqrt{r^4 + r^2 + m^2}$, $r > 0$, and $0 < m \leq 1/2$.

5 Conditional theorem

Theorem 1. *Assume Scalar Lemma 1. Then*

$$\text{aa}_{1/2}^1(2) = 1.$$

Consequently, under the same scalar lemma, Question 3 of [1] has a negative answer.

Proof. By the tree reduction recalled above, it suffices to prove the estimate for an arbitrary two-leaf metric tripod with edge lengths $u, v, w > 0$. We work with $p = 1/2$ and use the notation of Lemma 1.

Define

$$\lambda_1 = \frac{V + W - C}{A + B - C}, \quad \lambda_2 = \frac{W - B\lambda_1}{C}, \quad \lambda_3 = \frac{V - A\lambda_1}{C}.$$

The denominator $A + B - C$ is positive, and $V + W - C \geq 0$. The inequalities $\lambda_2, \lambda_3 \geq 0$ follow from concavity of $t^{1/2}$: for instance, if $C > A$, the increment $C - A$ is a terminal subinterval increment of $t^{1/2}$ inside the larger increment $C - V$, and hence

$$\frac{C - A}{C - V} \leq 1 \leq \frac{B}{W};$$

this is equivalent to $\lambda_2 \geq 0$. The other cases and $\lambda_3 \geq 0$ are symmetric or immediate. Directly from the definitions,

$$\lambda_1 + \lambda_2 + \lambda_3 = 1, \quad \lambda_1 A + \lambda_3 C = V, \quad \lambda_1 B + \lambda_2 C = W.$$

Scalar Lemma 1 is equivalent, after substituting the above value of λ_1 , to

$$\lambda_2 A + \lambda_3 B \leq U.$$

Now average the three three-point tree expressions N_1, N_2, N_3 :

$$\begin{aligned} \lambda_1 N_1 + \lambda_2 N_2 + \lambda_3 N_3 &= X(\lambda_1 A + \lambda_3 C) + Y(\lambda_1 B + \lambda_2 C) + Z(\lambda_2 A + \lambda_3 B) \\ &\leq XV + YW + ZU = D. \end{aligned}$$

Since the λ_i are nonnegative and sum to one,

$$\min\{N_1, N_2, N_3\} \leq \lambda_1 N_1 + \lambda_2 N_2 + \lambda_3 N_3 \leq D.$$

Thus the canonical inclusion $\mathcal{F}_p(\{0, x, y\}) \rightarrow \mathcal{F}_p(\{0, z, x, y\})$ has inverse norm at most 1 for every two-leaf metric tripod. The inverse norm is always at least 1, because the inclusion map is contractive. Hence every such tripod has amenability constant exactly 1. The source paper's reduction to two-leaf tripods gives $aa_{1/2}^1(2) = 1$.

Since Question 3 asks whether $aa_p^1(2) > 1$ for every $p \in (0, 1)$, the identity at $p = 1/2$ gives a negative answer, conditional on Lemma 1. $\square$

6 Verification notes

The included Python script performs deterministic log-grid and random checks of Scalar Lemma 1, the nonnegativity of the three weights, and the normalized Euclidean form. These checks are evidence only; they are not used as a proof.

The literature check was bounded. Local run indexes were searched for arXiv:2311.02920, $aa_p^1(2)$, $aa_p(2)$, and two-point p -amenability terms. Web searches for exact variants of these phrases found the source paper and the later broad embedding paper arXiv:2603.28193, but no answer to the sharp two-point $q = 1$ question.

7 Limitations

This packet is not a promoted counterexample. The proof of Theorem 1 is complete only after Scalar Lemma 1. Human review should either prove Lemma 1, find a counterexample to it, or replace it by a different certificate for $\min\{N_1, N_2, N_3\} \leq D$.

References

- [1] M. Cuth and T. Raunig, *Canonical embedding of Lipschitz-free p -spaces*, arXiv:2311.02920, 2023.
- [2] F. Albiac and J. L. Ansorena, *Lipschitz extensions into p -Banach spaces, and canonical embeddings of Lipschitz-free p -spaces for $0 < p < 1$* , arXiv:2603.28193, 2026.